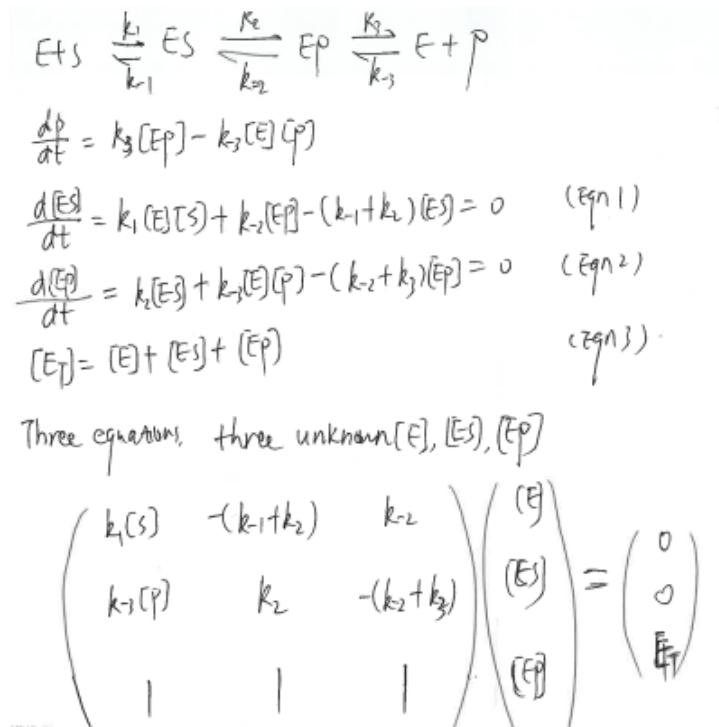


Reversible enzyme reactions, how to improve the net reaction rate?



% Here is how to solve the above equation

```
syms k1 k_minus_1 k2 k_minus_2 k3 k_minus_3 E ES EP S P Et;
```

```
>> eqn1 = k1*E*S + k_minus_2*EP -(k_minus_1 + k2)*ES ==0;
```

```
>> eqn2 = k2*ES + k_minus_3*E*P - (k_minus_2 + k3)*EP ==0;
```

```
>> eqn3 = E + ES + EP == Et;
```

```
>> latex(eqn1)
```

```
>> sol = solve([eqn1, eqn2, eqn3], [E, ES, EP])
```

```
'EP k_{-2} - ES (k_2 + k_{-1}) + E S k_1 = 0'
```

```
'ES k_2 - EP (k_3 + k_{-2}) + E P k_{-3} = 0'
```

```
'EP + ES + E = Et'
```

```
>> latex(sol.E)
```

$$\frac{Et k_2 k_3 + Et k_3 k_{-1} + Et k_{-1} k_{-2}}{k_2 k_3 + k_3 k_{-1} + k_{-1} k_{-2} + P k_2 k_{-3} + P k_{-1} k_{-3} + P k_{-2} k_{-3} + S k_1 k_2 + S k_1 k_3 + S k_1 k_{-2}}$$

```
>> latex(sol.EP)
```

$$\frac{Et (P k_2 k_{-3} + P k_{-1} k_{-3} + S k_1 k_2)}{k_2 k_3 + k_3 k_{-1} + k_{-1} k_{-2} + P k_2 k_{-3} + P k_{-1} k_{-3} + P k_{-2} k_{-3} + S k_1 k_2 + S k_1 k_3 + S k_1 k_{-2}}$$

% Here is the net rate of the reversible enzyme reaction

```
>> eqn4 = k3*sol.EP - k_minus_3*sol.E*P
```

>> simplify(eqn4)

ans =

$$-(E_t(P \cdot k_{\text{minus}_1} \cdot k_{\text{minus}_2} \cdot k_{\text{minus}_3} - S \cdot k_1 \cdot k_2 \cdot k_3)) / (k_2 \cdot k_3 + k_3 \cdot k_{\text{minus}_1} + k_{\text{minus}_1} \cdot k_{\text{minus}_2} + P \cdot k_2 \cdot k_{\text{minus}_3} + P \cdot k_{\text{minus}_1} \cdot k_{\text{minus}_3} + P \cdot k_{\text{minus}_2} \cdot k_{\text{minus}_3} + S \cdot k_1 \cdot k_2 + S \cdot k_1 \cdot k_3 + S \cdot k_1 \cdot k_{\text{minus}_2})$$

>> latex(ans)

$$-\frac{E_t(P k_{-1} k_{-2} k_{-3} - S k_1 k_2 k_3)}{k_2 k_3 + k_3 k_{-1} + k_{-1} k_{-2} + P k_2 k_{-3} + P k_{-1} k_{-3} + P k_{-2} k_{-3} + S k_1 k_2 + S k_1 k_3 + S k_1 k_{-2}}$$

% Here is how to simplify the net rate v, Matlab may also do this

$$= \frac{E_t(k_1 k_2 k_3 S - k_{-1} k_{-2} k_{-3} P)}{k_2 k_3 + k_3 k_{-1} + k_{-1} k_{-2} + k_{-3} P (k_2 + k_{-1} + k_{-2}) + k_1 S (k_2 + k_3 + k_{-2})}$$

$$\stackrel{110}{=} \frac{E_t \left(\frac{k_1 k_2 k_3 S - k_{-1} k_{-2} k_{-3} P}{k_2 k_3 + k_3 k_{-1} + k_{-1} k_{-2}} \right)}{1 + \frac{P k_{-3} (k_2 + k_{-1} + k_{-2}) + S k_1 (k_2 + k_3 + k_{-2})}{k_2 k_3 + k_3 k_{-1} + k_{-1} k_{-2}}}$$

let $\frac{k_2 k_3 + k_3 k_{-1} + k_{-1} k_{-2}}{k_{-3} (k_2 + k_{-1} + k_{-2})} = K_p$; $\frac{k_2 k_3 + k_3 k_{-1} + k_{-1} k_{-2}}{k_1 (k_2 + k_3 + k_{-2})} = K_s$

$$V = \frac{E_t \left(\frac{S}{K_s} \cdot \frac{k_2 k_3}{k_2 + k_3 + k_{-2}} - \frac{P}{K_p} \frac{k_{-1} k_{-2}}{k_2 + k_{-1} + k_{-2}} \right)}{1 + \frac{P}{K_p} + \frac{S}{K_s}}; \quad \frac{k_2 k_3}{k_2 + k_3 + k_{-2}} = k^+; \quad \frac{k_{-1} k_{-2}}{k_2 + k_{-1} + k_{-2}} = k^-;$$

($k^+ E_t = V_m^+$
 $k^- E_t = V_m^-$)

$$V = \frac{E_t \left(\frac{S}{K_s} \frac{k_2 k_3}{k_2 + k_3 + k_{-2}} - \frac{P}{K_p} \frac{k_{-1} k_{-2}}{k_2 + k_{-1} + k_{-2}} \right)}{1 + \frac{S}{K_s} + \frac{P}{K_p}}$$

$$\text{let. } k_f = \frac{k_2 k_3}{k_2 + k_3 + k_{-2}}; \quad k_r = \frac{k_{-1} k_{-2}}{k_2 + k_{-1} + k_{-2}}$$

$$V = \frac{k_f E_t \cdot \frac{S}{K_s} - k_r E_t \cdot \frac{P}{K_p}}{1 + \frac{S}{K_s} + \frac{P}{K_p}}$$

$$= \frac{V_m^+ \frac{S}{K_s} - V_m^- \frac{P}{K_p}}{1 + \frac{S}{K_s} + \frac{P}{K_p}}$$

when $P=0$,

$$V = \frac{V_m^+ \frac{S}{K_s}}{1 + \frac{S}{K_s}} = \frac{V_m^+ S}{K_s + S}$$

Discussion

1. If we want to increase the net reaction rate, you need to increase total enzyme E_t . We can achieve this by gene overexpression;
2. To increase the net rate, we can also increase k_f , to achieve this, we need to increase k_2 or k_3 , but decrease k_{-2} . Which means the enzyme-substrate complex needs to be quickly converted to EP complex and EP complex is quickly converted to product P. To decrease k_{-2} , we need to lower down the propensity from EP to ES. This can be achieved by using protein engineering to mutate the ES binding pocket, increase the stability of EP complex and improve the catalytic activity at the catalytic center to facilitate ES to EP conversion.
3. To increase net rate, we can also decrease k_r , this can be achieved by decreasing k_{-1} or k_{-2} , but increase k_2 . We need to design the EP and ES binding pocket which restricts the back flow from EP to ES and ES to E. To increase k_2 , we need to improve the turnover rate of ES to EP.
4. We may also decrease K_s but increase K_p , which means we need to create the situation to favor the formation of ES complex, but destabilizes EP, allowing product P quickly exit from the EP complex.
5. The reverse rate is correlated with the forward rate by the equilibrium constant K . That's why the reverse rate can be re-written into the form of Gibbs energy change (listed here). You can refer to the Process review paper or the Nature chemical biology paper. With these constraints, you may evaluate how temperature may affect the net reaction rate.

$$\Delta_r G^{\circ'} = -RT \ln K$$

$$\Delta_r G' = \Delta_r G^{\circ'} + RT \ln Q$$

$$v = v_m^+ \frac{\frac{[S]}{k_s}}{1 + \frac{[S]}{k_s} + \frac{[P]}{k_p}} \left(1 - e^{\frac{\Delta_r G}{RT}} \right) = v_m^+ \frac{\frac{[S]}{k_s}}{1 + \frac{[S]}{k_s} + \frac{[P]}{k_p}} \left(1 - \frac{Q}{K} \right)$$